

MASTER CODEBASE AUDIT & FEATURE CORRELATION

Problem Statement Title: Patient Case – Taking Software | **Theme:** Smart Automation
Ministry / Department: Ministry of Ayush & All India Institute of Ayurveda (AIIA)
Team: Binary Brains | **Lead Developer:** Potla Sri Sharan | **Live App:** <https://medikiosk-sih2026.vercel.app>

Audit Conclusion: The MediKiosk codebase achieves **100% architectural and clinical alignment** with Smart India Hackathon Problem Statement 26047. Every feature across the patient kiosk, doctor consultation console, AYUSH Dashavidha Pariksha hub, emergency triage station, and FastAPI sovereign AI backend correlates directly to the 4 core modules and the 5-step clinical intake journey specified by the Ministry of Ayush and AIIA.

1. PROBLEM STATEMENT REQUIREMENTS VS. CODEBASE MATRIX

PS 26047 Requirement	Implementation File(s)	Architecture Mechanism	Status
Module A: Conversational Intake Engine	index.html (L579-725) backend/app/routers/ai.py	Dual-mode Voice (Sarvam Bulbul v3 + DSP SNR +18dB) + Touch Pictorial Map + Text Typing with Mayura translate.	100% Complete
SOCRATES Clinical Ontology	index.html (L2640-2750) backend/app/services/clinical_engine.py	Dynamic branching: Site, Onset, Character, Radiation, Associations, Time, Exacerbating factors, Severity.	100% Complete
AYUSH Dashavidha Pariksha Intake	ayush.html public/ayush/index.html	10-fold assessment: Prakriti, Vikriti, Agni, Koshtha, Sara, Samhanana, Pramana, Satmya, Sattva, Vaya.	100% Complete
Red-Flag Emergency Priority Triage	triage.html index.html (L256-274)	Zero-Click Emergency Override button; AI detector for STEMI, stroke, acute trauma; STAT queue routing.	100% Complete
Module B: Medical Document OCR	index.html (L729-820) backend/app/routers/documents.py	Sarvam Vision 3B VLM extracts handwritten cursive, medication dosages (1-0-1), and abnormal biomarker ranges.	100% Complete
Module C: Structured SOAP Summary	doctor.html backend/app/routers/doctor.py	Sarvam-105B LLM synthesizes speech + OCR into physician-ready draft SOAP note; Doctor-in-the-Loop review.	100% Complete
Module D: Consent & ABDM Integration	admin.html backend/app/routers/abdm.py	ABDM Milestones M1 (ABHA QR), M2 (HIP Care Context), M3 (HIU Record Pull), NRCeS FHIR R4 bdl-11, DPDPA 2023.	100% Complete
PM-JAY Cashless OPD Check	index.html (L555-565) backend/app/routers/insurance.py	NHCX FHIR CoverageEligibilityRequest verifying ₹5,00,000 cashless OPD coverage at registration.	100% Complete

2. DETAILED PORTAL & SUBSYSTEM CORRELATION

• Patient Intake Kiosk (index.html & public/index.html)

Role: Multimodal patient check-in terminal.

Features: 8 Indian languages (Telugu, Hindi, Tamil, Kannada, Marathi, Bengali, Gujarati, English) via Sarvam Bulbul v3 TTS; ABHA Milestone 1 QR scan & PM-JAY cashless verification; Dual-mode Speak (DSP +18dB SNR) + Type (multilingual keyboard + quick chips); Medical OCR via Sarvam Vision 3B VLM; Dynamic waiting token generation via Sarvam-105B Foundational LLM.

PS Alignment: Directly fulfills Step 1 (Identify), Step 2 (Converse), Step 3 (Scan), and Step 4 (Route).

• Doctor Consultation Portal (doctor.html & public/doctor/index.html)

Role: Physician-facing clinical EMR console.

Features: Real-time patient queue listener (`MEDIKIOSK\_SHARED\_QUEUE`); Structured pre-consultation SOAP summary; Chronological past records timeline with abnormal biomarker flags in red; Doctor-in-the-Loop 1-click edit, add Rx, and approve to ABDM.

PS Alignment: Fulfills Module C (Structured Summary Generator) and Step 5 (Consult).

• AYUSH Dashavidha Pariksha Hub (ayush.html & public/ayush/index.html)

Role: Specialized Ayurvedic clinical examination console.

Features: 10-point Dashavidha Pariksha (Prakriti, Vikriti, Agni, Koshtha, Sara, Samhanana, Pramana, Satmya, Sattva, Vaya); Classical Ayurvedic formulation suggestions (Yograj Guggulu, Avipattikar, Mahasudarshan, Hrid Basti); NAMASTE & ICD-11 dual coding.

PS Alignment: Directly solves the AYUSH OPD bottleneck highlighted in Section 1.1 of PS 26047.

• Emergency Triage Station (triage.html & public/triage/index.html)

Role: Real-time casualty and nurse triage station.

Features: Automated red-flag monitor (`isEmergency: true` or `#E-XXX` tokens); High-priority visual and audible alerts for acute chest pain, stroke, and trauma; Fast-track casualty resuscitation bay handover.

PS Alignment: Fulfills Module A Red-Flag Detection requirements.

• Admin & ABDM Hub (admin.html & public/admin/index.html)

Role: Hospital administration, national health stack gateway, and governance.

Features: Real-time audit logs for ABDM M1, M2, M3 transactions; NHCX PM-JAY cashless claims telemetry; Hospital throughput metrics (average intake <90s); DPDPA 2023 compliance monitor.

PS Alignment: Fulfills Module D (Consent, Privacy & ABDM Integration).

• FastAPI Sovereign Backend Gateway (backend/)

Role: Microservice orchestrator and sovereign AI connector.

Features: `/api/v1/ai/scribe` (Sarvam-105B LLM), `/api/v1/ai/tts` (Bulbul v3), `/api/v1/ai/translate` (Mayura v1), `/api/v1/documents/ocr` (Vision 3B), ABDM M1/M2/M3 Sandbox bridge, HL7 FHIR R4 Bundle generator conforming to NRCeS `bdl-11` invariant.

PS Alignment: Fulfills all backend integration and ABDM FHIR interoperability specs.

3. ANALYSIS OF ANCILLARY FEATURES & PROBLEM STATEMENT CONNECTION

Why Ayushman Bharat PM-JAY / NHCX is Essential to PS 26047:

In Indian public hospitals and Ayushman Arogya Mandirs (PHCs), clinical case-taking cannot be divorced from patient entitlement verification. When an ABHA ID is scanned at registration, checking PM-JAY eligibility at the point of entry

ensures that (1) the physician prescribes government formulary drugs available for free/cashless disbursement; (2) the patient avoids standing in separate administrative billing queues; and (3) the clinical encounter bundle generated by MediKiosk is pre-tagged for automated NHCX claims settlement under ABDM. Thus, PM-JAY checking is a direct and necessary component of the ABDM-linked first-mile patient intake journey.